

NTRK3 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NTRK3 gene fusions from the msk_impact_50k_2026 study, covering 58 fusion events. 55/58 fusions (94.8%) were in-frame. 55/58 fusions (94.8%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.0695489$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for NTRK3.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 58 analyzed fusion events, identifying 4 distinct fusion partner genes. The most frequent partner was ETV6, observed in 55 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 54/55 (98.2%) of in-frame fusions retained the domain, $p=0.0695489$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 54/55 (98.2%) of in-frame fusions disrupted the domain, $p=0.0695489$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.663366.

Cutpoint detection

Cutpoint detection scanned 57 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 155 aa (permutation-corrected $p=0.036963$, statistically significant at $\alpha=0.05$). This is -5 aa from the nearest configured domain boundary at 160 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=55$) and "not_retained" ($n=2$). For Frame_status, retained: 54/54 (100.0%, 95% CI 93.4%-100.0%); not_retained: 1/2 (50.0%, 95% CI 9.5%-90.5%). Welch's t-test comparing Tumor_variant_count between groups gave means of 74.1455 vs 7, $p=5.97753e-12$ (statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (4 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: NTRK3 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ETV6	55	0.9482758620689655	0.20043213737826426	0.017824657108181614	0.0	0.5613244116887826	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.39428816555577034	1
DOT1L	1	0.017241379310344827	0.20043213737826426	0.017824657108181614	1.0	0.5613244116887826	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2649778207281841	2
VPS39	1	0.017241379310344827	0.20043213737826426	0.017824657108181614	None	0.5613244116887826	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.13526802438609895	3
ZNF710	1	0.017241379310344827	0.20043213737826426	0.017824657108181614	0.11159235668789813	0.5613244116887826	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.13171667423136882	4

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
155	0	55	1	1	0.0	0.03508771929824562	1.4548448600085102
466	14	41	2	0	0.0	0.07518796992481205	1.1238516409670856
472	15	40	2	0	0.0	0.08521303258145364	1.069493978644493

domain_disruption tables

frame_domain_contingency_table

54	1
1	1

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf01462	Leucine rich repeat N-terminal domain	57	2	0	55	0.0
auto_disruption_pf13855	Leucine rich repeat	57	1	1	55	0.016290726817042606
auto_disruption_pf07679	Immunoglobulin I-set domain	57	1	0	56	0.0

domain_retention tables

frame_domain_contingency_table

54	1
1	1

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	57	55	0	2	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
14	55	58	0.9482758620689655

frequency tables

Partner_gene_counts

Partner_gene	Event_count
DOT1L	1
ETV6	55
VPS39	1
ZNF710	1

Per-event results (results.tsv)

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-001 3664-T02-I M5-1	P-0013 664-T0 2-IM5		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ pr ime	885581 91	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15) (p13.2;q26.1)(chr12:g.12032 077::chr15:g.88558191) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-006 1265-T01-I M7-2	P-0061 265-T0 1-IM7		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ pr ime	885541 22	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 33175::chr15:g.88554122) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-005 8408-T01-I M6-3	P-0058 408-T0 1-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ pr ime	884921 56	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 35760::chr15:g.88492156) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-005 6156-T02-I M7-4	P-0056 156-T0 2-IM7		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ _pr ime	886242 74	13	466	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 11957::chr15:g.88624274) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-005 5703-T01-I M6-5	P-0055 703-T0 1-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ _pr ime	885059 80	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 24785::chr15:g.88505980) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-004 2503-T01-I M6-6	P-0042 503-T0 1-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ _pr ime	885560 67	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 32802::chr15:g.88556067) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040355-T01-IM6-7	P-0040355-T01-IM6		ETV6-NTKR3	ETV6	in-frame	threonine_prime	88497191	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12036111::chr15:g.88497191) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0037610-T01-IM6-8	P-0037610-T01-IM6		ETV6-NTKR3	ETV6	in-frame	threonine_prime	88594532	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) rearrangement: t(12;15)(p13.2;q25.3)(chr12:g.12014663::chr15:g.88594532) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0035767-T06-IM7-9	P-0035767-T06-IM7		ETV6-NTKR3	ETV6	in-frame	threonine_prime	88505787	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-003 5767-T04-I M6-10	P-0035 767-T0 4-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ _pr im e	885057 87	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epea t N-te rmin al do main; Leuci ne rich r epea t; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 29143::chr15:g.88505787) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 3921-T05-I M7-11	P-0033 921-T0 5-IM7		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ _pr im e	885165 34	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epea t N-te rmin al do main; Leuci ne rich r epea t; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 31161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 3921-T04-I M6-12	P-0033 921-T0 4-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ _pr im e	885165 34	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epea t N-te rmin al do main; Leuci ne rich r epea t; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 31161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 3921-T02-IM6-13	P-0033 921-T02-IM6		ETV6-NT RK3	ETV6	in-frame	threonine_prime	88516534	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 2148-T04-IM6-14	P-0032 148-T04-IM6		ETV6-NT RK3	ETV6	in-frame	threonine_prime	88522437	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12037217::chr15:g.88522437) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-006 4964-T01-IM7-15	P-0064 964-T01-IM7		ETV6-NT RK3	ETV6	in-frame	threonine_prime	88564682	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12032310::chr15:g.88564682) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 3664-T04-IM6-16	P-0013 664-T04-IM6		ETV6-NTRK3	ETV6	in-frame	three_prime	88558191	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032077::chr15:g.88558191) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-005 8230-T02-IM7-19	P-0058 230-T02-IM7		NTRK3-DOT1L	DOT1L	out-of-frame	five_prime	88683244	5	155	True	lost	0.0	False	Leucine rich repeat N-terminal domain	Protein tyrosine and serine/threonine kinase; Immunoglobulin I-set domain	Leucine rich repeat	NTRK3 (NM_001012338) - DOT1L (NM_032482) rearrangement: t(15;19)(q25.3;p13.3)(chr15:g.88683244::chr19:g.2227604) Protein Fusion: out of frame {NTRK3:DOT1L}	NA
EVT-P-001 3196-T03-IM6-20	P-0013 196-T03-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88501357	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-000 2978-T06-I M6-21	P-0002 978-T0 6-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	885065 71	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88 506571) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-001 0668-T01-I M5-22	P-0010 668-T0 1-IM5		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	885132 82	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2; q25.3)(chr12:g.12029109::chr 15:g.88513282) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-001 4872-T02-I M6-23	P-0014 872-T0 2-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	885480 29	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2; q25.3)(chr12:g.12028946::chr 15:g.88548029) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-001 4872-T03-I M6-24	P-0014 872-T0 3-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	885480 29	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epea t N-te rmin al do main; Leuci ne rich r epea t; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2; q25.3)(chr12:g.12028946::chr 15:g.88548029) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 3196-T04-I M6-25	P-0013 196-T0 4-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	885013 57	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epea t N-te rmin al do main; Leuci ne rich r epea t; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3) (chr12:g.12036072::chr15:g.8 8501357) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 6507-T01-I M6-26	P-0016 507-T0 1-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	884888 20	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epea t N-te rmin al do main; Leuci ne rich r epea t; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88 488820) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-001 6507-T02-I M6-27	P-0016 507-T0 2-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	884888 20	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-14 fused with NTRK3 exons 6-20 including the kinase domain): t(12;15)(p 13.2;q25.3)(chr12:g.12023709 ::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 6507-T03-I M6-28	P-0016 507-T0 3-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	884888 20	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20: t(12;15)(p13.2;q25.3)(chr12:g.1202370 9::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 6507-T04-I M6-29	P-0016 507-T0 4-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	884888 20	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20): t(12;15) (p13.2;q25.3)(chr12:g.12023 709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0002671-T01-IM3-30	P-0002671-T01-IM3		NTRK3-ETV6	ETV6	in-frame	threepri	88505784	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		NA Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0018060-T01-IM6-31	P-0018060-T01-IM6		NTRK3-ETV6	ETV6	in-frame	threepri	88514531	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.88514531) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0021974-T01-IM6-32	P-0021974-T01-IM6		NTRK3-ETV6	ETV6	in-frame	threepri	88550500	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028417::chr15:g.88550500) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 3196-T02-IM6-33	P-0013 196-T02-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88501357	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin-set domain		ETV6 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-002 5507-T01-IM6-34	P-0025 507-T01-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88544573	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin-set domain		ETV6 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12024078::chr15:g.88544573) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 5987-T01-IM6-35	P-0025 987-T01-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88613114	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin-set domain		ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12028942::chr15:g.88613114) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 0562-T01-IM5-36	P-0010 562-T01-IM5		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88581658	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin-like domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 14-20): t(12;15)(p13.2;q25.1)(chr12:g.12028903::chr15:g.88581658) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-002 6390-T01-IM6-37	P-0026 390-T01-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88651131	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin-like domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12016863::chr15:g.88651131) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 6696-T01-IM6-38	P-0026 696-T01-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88657925	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin-like domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12007967::chr15:g.88657925) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027006-T01-IM6-39	P-0027006-T01-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88512343	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12032352::chr15:g.88512343) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0027006-T03-IM6-40	P-0027006-T03-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88512343	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.88514531) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0027006-T04-IM6-41	P-0027006-T04-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88512343	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352::chr15:g.88512343) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_ retained_ fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-000 2978-T07-I M6-42	P-0002 978-T0 7-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	885065 71	14	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88 506571) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-000 2978-T02-I M5-43	P-0002 978-T0 2-IM5		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	885065 71	14	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		NA Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-002 6286-T01-I M6-44	P-0026 286-T0 1-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	886114 76	13	466	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15) (p13.2;q25.3)(chr12:g.120303 95::chr15:g.88611476) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-000 2978-T03-I M5-45	P-0002 978-T0 3-IM5		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	885065 71	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 5 fused to NTRK3 exons 15 to 20) : t(12; 15)(p13.2;q25.3)(chr12:g.120 28576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-001 0264-T01-I M5-46	P-0010 264-T0 1-IM5		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	885882 55	13	466	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused in frame with NTRK3 exons 13-20): t(12;15)(p13.2;q25.3)(chr12:g.12014962::chr15:g.88 588255) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-000 7911-T02-I M5-47	P-0007 911-T0 2-IM5		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr im e	885924 30	13	466	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epeat N-te rmin al do main; Leuci ne rich r epeat ; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 4 fused with NTRK3 exons 13-20): t(12;15) (p13.2;q25.3)(chr12:g.120071 42::chr15:g.88592430) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0033921-T03-IM6-48	P-0033921-T03-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88516534	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0007772-T03-IM5-49	P-0007772-T03-IM5		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88600844	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0007772-T02-IM5-50	P-0007772-T02-IM5		NTRK3-ETV6	ETV6	in-frame	five_prime	88600844	12	466	True	lost	0.0	False	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain	Protein tyrosine and serine/threonine kinase		ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) Protein fusion: in frame (NTRK3-ETV6)	NA

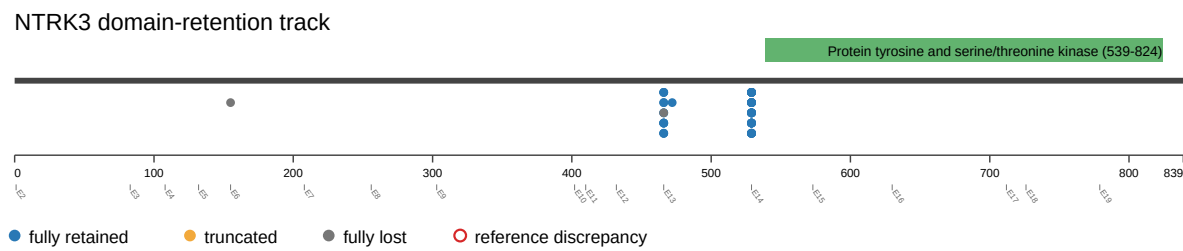
event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-003 5552-T01-I M6-51	P-0035 552-T0 1-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr ime	886512 51	13	466	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epea t N-te rmin al do main; Leuci ne rich r epea t; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15) (p13.2;q25.3)(chr12:g.120209 63::chr15:g.88651251) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 5767-T01-I M6-52	P-0035 767-T0 1-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr ime	885057 87	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epea t N-te rmin al do main; Leuci ne rich r epea t; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 29143::chr15:g.88505787) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-000 7772-T01-I M5-53	P-0007 772-T0 1-IM5		NTR K3-E TV6	ETV 6	in-fr ame	thre e_ _pr ime	886008 44	13	466	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Leuci ne rich r epea t N-te rmin al do main; Leuci ne rich r epea t; Im mun oglob ulin I-set dom ain		ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(1 2;15)(p13.2;q25.3)(chr12:g.12 020257::chr15:g.88600844) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006480-T01-IM5-54	P-0006480-T01-IM5		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88498145	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029548::chr15:g.88498145) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0002978-T04-IM5-55	P-0002978-T04-IM5		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88506571	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0002978-T05-IM6-56	P-0002978-T05-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88506571	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA

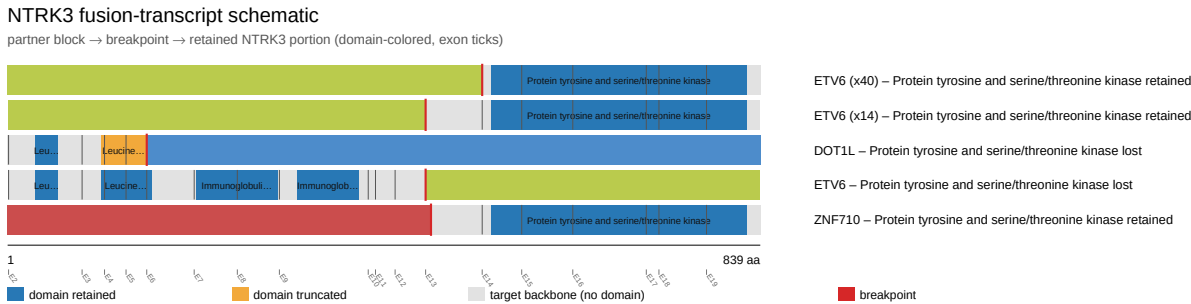
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004737-T01-IM5-57	P-0004737-T01-IM5		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88632786	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001012338) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 13 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12006809::chr15:g.88632786) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0023257-T01-IM6-58	P-0023257-T01-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88662190	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		ETV6 (NM_001012338) fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 14 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12014998::chr15:g.88662190) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0010556-T01-IM5-70	P-0010556-T01-IM5		NTRK3-ZNF710	ZNF710	unknown	threonine_prime	88576258	13	472	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Leucine rich repeat N-terminal domain; Leucine rich repeat; Immunoglobulin I-set domain		NTRK3 (NM_001012338) - ZNF710 (NM_198526) rearrangement: c.-28-10227:ZNF710_c.1415:NTRK3inv Protein fusion: mid-exon (ZNF710-NTRK3)	NA

Figures

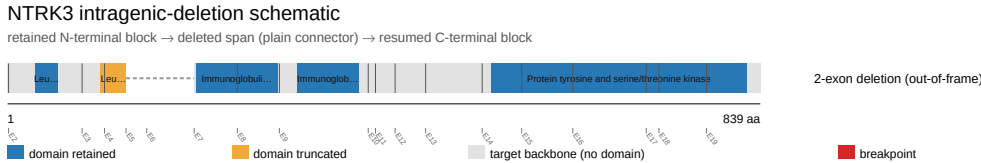
domain retention outliers



fusion schematic



intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

